

AI 3603 Artificial Intelligence: Principles and Techniques

Homework 2 Deadline: Nov. 18th 22:00 p.m.

Adhere to the Code of Academic Integrity. You may discuss background issues and general strategies with others and seek help from course staff, but the implementations that you submit must be your own. In particular, you may discuss general ideas with others but you may not work out the detailed solutions with others. It is never OK for you to see or hear another student's code and it is never OK to copy code from published/Internet sources. Moss (Measure Of Software Similarity) will be used for determining the similarity of programs to detect plagiarism in the class (<https://theory.stanford.edu/~aiken/moss/>) If you encounter some difficulties or feel that you cannot complete the assignment on your own, discuss with your classmates in Discussion forum on Canvas, or seek help from the course staff.

You are required to complete this homework *individually*. Please submit your assignment following the instructions summarized in Section 7.

1 Reinforcement Learning in Cliff-walking Environment

In this assignment, you will implement Reinforcement Learning agents to find a safe path to the goal in a grid-shaped maze. The agent will learn by trial and error from interactions with the environment and finally acquire a policy to get as high as possible scores in the game.

1.1 Game Description

Suppose a 12×4 grid-shaped maze in Fig. 1. The bottom left corner is the starting point, and the bottom right corner is the exit. You can move upward, downward, leftward, and rightward in each step. You will stay in place if you try to move outside the maze. You are asked to reach the goal through the safe region and avoid falling into the cliff. Reaching the exit terminates the current episode, while falling into the cliff gives a reward -100 and return to the starting point. Every step of the agent is given a living cost (-1).

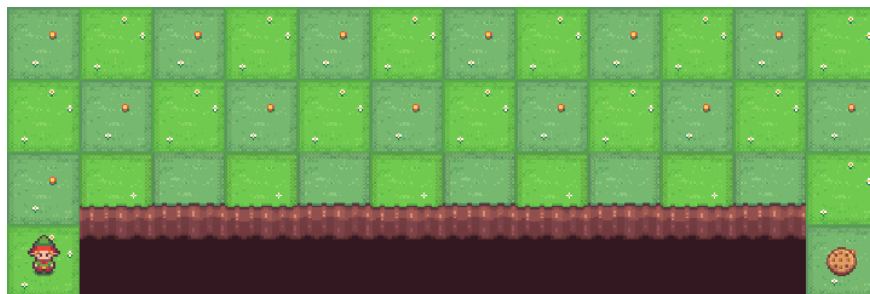


Figure 1: The cliff-walking environment

The state space and action space are briefly described as follows:

State: s_t is an integer, which represents the current coordinate (x, y) of the agent.

Action: $a_t \in \{0, 1, 2, 3\}$, where four integers represent four moving directions respectively.

1.2 Implement Sarsa, Q-Learning, and Dyna-Q

You are asked to implement agents based on Sarsa, Q-Learning, and dyna-Q algorithms. Please implement the agents in `agent.py` and complete the training process in `cliff_walk_sarsa.py`, `cliff_walk_qlearning.py`, and `cliff_walk_dyna_q.py` respectively. An agent with a random policy is provided in the code. You can learn how to interact with the environment through the demo and then write your own code.

Hint: Take `cliff_walk_sarsa.py` as an example:

- Line 27: more parameters need to be utilized to construct the agent, such as learning rate, reward decay γ , ε value, and ε -decay schema.
- Line 47: the agent needs to be provided with some experience for learning.

Hint: In `agent.py`:

- You need to implement ε -greedy with `$\varepsilon$  value decay` in the `choose_action` function.
- Functions given in the template need to be completed. You can also add other utility functions as you wish in the agent classes.

Hint: You should **keep a balance between exploration and exploitation** by tuning ε value carefully. In the early stage of the training, exploration should be encouraged to get familiar with the environment. With the advancement of training, exploration may be reduced for the convergence of the policy.

1.3 Result Visualization and Analysis

Result Visualization: You are required to visualize the training process and the final result according to the following requirements:

1. Plot the episode reward during the training process.
2. Plot the ε value during the training process.
3. Visualize the final paths found by the intelligent agents after training.

Result Analysis: You are required to analyze the learning process based on the experiment results according to the following requirements:

1. You may find that there exists a little difference between the paths found by Sarsa and Q-learning. Please describe the difference in the report and analyze the reason in detail.
2. Analyze the training efficiency between model-based RL (dyna-Q) and model-free algorithms (Sarsa or Q-learning). Please describe the difference in the report and analyze the reason in detail.

2 Deep Reinforcement Learning

In this part, you will implement a DQN agent to control a lunar lander. You are asked to read and running the given code to train your intelligent lander agent. The performance of the agent may not be satisfactory and you have to tune it to get higher scores.

2.1 Game Description

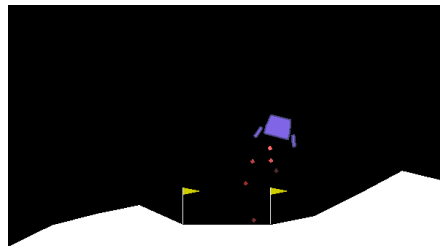


Figure 2: The lunar lander environment in this assignment

This task is a classic rocket trajectory optimization problem. As shown in Fig. 2, you are required to control the the space-ship and land between the flags smoothly.

In this assignment, you are required to train a DQN agent in “LunarLander-v2” gym environment. The definitions of state and action are given as follows:

State(Array): The state s_t is an 8-dimensional vector: the coordinates of the lander in x & y, its linear velocities in x & y, its angle, its angular velocity, and two booleans that represent whether each leg is in contact with the ground or not.

Action(Integer): There are four discrete actions available: do nothing, fire left orientation engine, fire main engine, and fire right orientation engine.

More details of this gym environment is given in the documents of gym¹. However, information given in this file is sufficient for this assignment.

2.2 Read and Analyze the Deep Q-Network Implementation

In this section, a complete DQN implementation `dqn.py` is given for the lunar lander task. You are required to read the code and understand the DQN training process. You are required to **write comments** in the code to point out the function or your understanding of each part of the code. Please fill in every ““**comments:** ””” (cf. `dqn.py`) with your understanding of the code.

2.3 Train and Tune the Agent

In this section, you are required to train the DQN agent. Please show your training process (such as learning curve of episode return) and the training result (such as the game video of the final model) in the report.

The performance of the given code may be not satisfying. You need to tune the agent to get higher scores within 500000 time-steps or make the learning process more efficient. Here are some possible ways to improve the performance:

- **(Requested)** Tuning the hyper-parameters of the agent, especially gamma value, epsilon value, and epsilon decay schema.
- Tuning the structure of the Q network.
- Utilize multiple continuous frames of the game instead of one frame each time.
- Other ideas.

3 Improve Exploration Schema

There exists lots of other exploration strategies except ϵ -greedy. You are asked to find and learn one new exploration method, such as Upper Confidence Bound(UCB). Summarize the idea, pros, and cons of the new exploration method. Write your understanding in the report.

4 Installation

You can follow the tutorial in this section to install the environment on Linux or Windows, and we strongly recommend you to use Linux system.

4.1 Install Anaconda

Open the address <https://www.anaconda.com/distribution/> and download the installer of **Python 3.x version**(3.8 recommended) for your system.

¹https://www.gymnasium.dev/environments/box2d/lunar_lander/

4.2 Install Required Environment

After installing anaconda, open a Linux terminal and create an environment for Gym:

```
conda create python=3.8 --name gym
```

Then activate the environment

```
conda activate gym
```

Install gym and some dependencies

```
pip install gym==0.25.2
```

```
pip install gym[box2d]
```

```
pip install stable-baselines3==1.2.0
```

```
pip install tensorboard
```

Install pytorch: Please follow the instructions given on the pytorch website².

5 Code, Demo Video, and Report

Code: You can edit the code between “##### START CODING HERE #####” and “##### END CODING HERE #####”. Please **DON’T** modify other parts of the code.

Demo Video: Videos (optional) should be in .mp4 format and a 10MB max for a single file . You can compress/speed up the videos. We recommend recording videos utilizing gym wrappers: “env = gym.wrappers.RecordVideo(env, './video’)”. More information is given in the gym docs³. All the videos should be put into a folder called **videos**.

Report: Summarize the process and results of the homework.

6 Discussion and Question

You are encouraged to discuss your ideas, ask and answer questions about this homework. If you encounter any difficulty with the assignment, try to post your problem **on Canvas** for help. The classmates and the course staff will try to reply.

7 Submission instructions

1. Zip all your program files, experiment result, and report file **HW2_report.pdf** to a file named as **HW2_ID_name.zip**.
2. Upload the file to the homework 2 page on the Canvas.

²<https://pytorch.org/get-started/locally/>

³<https://www.gymnasium.dev/api/wrappers/>